P530/1
BIOLOGY
Paper 1
2026
3 hours

UGANDA TEACHERS' EDUCATION CONSULT (UTEC)

Uganda Advanced Certificate of Education

BIOLOGY

Paper 1

(Theory)

3 hours

INSTRUCTIONS TO CANDIDATES:

This examination paper consists of **two** *sections;* **A** *and* **B**. *It has* **six** *items.*

Section **A** *has* **two compulsory** *items.*

Section **B** *has* **two** *parts:* **I** *and* **II**. *Respond to* **one** *item from* **each** *part.*

Respond to **four** *items in* **all**.

All responses must be written in answer booklet(s) provided.

Any additional item(s) responded to will not be scored.

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SECTION A

Respond to **both** *items in this section*

Item 1

Due to the need to improve the treatment of patient with severe skin injuries and inherited disorders, Mulago Regional Tissue Engineering Centre established a tissue culture and molecular biology laboratory to develop artificial skin for treating severe burns and genetically compatible stem cells for transplantation.

However, experts observed that some cultured skin tissue wounds repaired slowly while others developed abnormal cells that could not be used for transplantation. In addition, many patients requiring stem-cell transplantation failed to obtain suitable donors because of poor genetic compatibility.

To improve treatment outcomes, scientists investigated factors affecting tissue repair, cell division and donor compatibility. **Tables 1** and **2** show the results obtained.

Table 1: Characteristics of cultured skin cells

Cell culture	**Culture A**	**Culture B**
Number of mitochondria per cell	235	118
Number of rough endoplasmic reticulum	162	74
Percentage of cells completing mitosis	94	56
Percentage wound closure after 7 days	96	57

Table 2: Characteristics of reproductive cells from potential stem-cell donors

Donor group	**Percentage of cells undergoing meiosis**	**Percentage of genetically different reproductive cells formed**	**Percentage of compatible stem-cell donor matches**
Group A	95	97	73
Group B	58	49	34

To improve tissue repair and increase the availability of compatible stem cells, the laboratory scientists proposed the following interventions:

(i) Correct defective genes in a patient's stem cells using gene editing before culturing them.

(ii) Continuously exposed cultured skin cells to high concentrations of growth factors (growth factors increase rate of DNA replication).

(iii) Recruit more volunteer stem cell donors without introducing gene editing.

Task:

(a) Analyze the information provided to explain the differences in tissue repair and stem-cell donor compatibility of groups A and B.

(b) Evaluate the effectiveness of the proposed interventions in improving tissue repair and stem cell transplantation.

Item 2

As part of the government's programme to increase production of medicines from natural tree extracts, National Drug Authority (NDA) established a new Moringa tree plantation using seeds. However, when the seeds were sown, only a few germinated. After transplanting the germinated seedlings, they introduced a cover crop to conserve soil moisture and suppress weed growth. Unlike the Moringa trees, the cover crop uses two carbon dioxide fixing enzymes during photosynthesis. During the hot season, the plantation manager observed that the Moringa trees were not growing at the expected rate. To establish the causes of stunted growth, an agricultural officer conducted a study to compare the photosynthetic performance of Moringa trees and cover crops and also investigated the factors responsible for the low percentage germination of Moring tree seeds.

Table 3: *Photosynthetic characteristics of the moringa trees and cover crops.*

Plant	**Activity of enzymes (%)**		**Soluble sugars produced (g plant^{-1} day^{-1})**
	Rubisco	**PEP Carboxylase**	
Moringa trees	58	Enzyme absent	20
Cover crop	36	89	34

The medicinal extracts are synthesized from photosynthetic sugars.

Table 4: *Hormonal changes associated with germination of moringa seeds.*

Storage condition	**Abscisic acid (µg g^{-1} seed)**	**Gibberellin (µg g^{-1} seed)**	**Amylase activity (%)**	**Germinating Seedlings (out of 100 seeds)**
Cool, moist	5	20	86	91
Hot, dry	17	8	38	42

To improve seed germination and synthesis of medicinal products, the agricultural officer recommended the following interventions:

(i) Treatment of all stored seeds with gibberellin immediately after harvesting.

(ii) Insert genes from the cover crop into moringa trees *(this allows them to concentrate carbon dioxide around RUBISCO).*

(iii) Re-locate the Moringa tree plantation to a highland area *(where temperatures are relatively low most of the time)*.

The plantation manager needs an evaluation on the effectiveness of each intervention and how to apply them.

Task:

(a) Analyse the causes of the differences in sugar production and seed germination.

(b) Evaluate the effectiveness of the recommended interventions and advice the plantation manager on how to apply them.

SECTION B:

Part I

Respond to **one** *item from this Part*

Item 3

A district hospital conducted a medical outreach in a Sub-County to investigate common health conditions affecting residents. During the outreach, Paul reported excessive thirst and passing out small volumes of concentrated urine after working under the hot sun. Sarah complained of becoming breathless after carrying heavy luggage. Blood and urine samples were collected from Paul and Sarah for laboratory analysis by the medical team.

During the same outreach, doctors examined John who had sustained a minor cut on the legs two days earlier. John's wound was warm, swollen and painful. Juliet, a resident in the area, complained of difficulty maintaining body balance while walking and her clinical examination results showed an infection that had damaged the semi-circular canals of the inner ear. **Table 5** shows the laboratory results conducted on different residents (Paul & Sarah).

Table 5

Physiological characteristic affected	***Paul***	***Sarah***	***Expected normal range***
Antidiuretic hormone levels (ADH) / Pg mL^{-1}	5.9	4.3	1.0 - 5.0
Urine Volume (L)	0.7	1.8	0.8 - 2.0
Oxygen Saturation %	98	95	95 - 100
Haemoglobin concentration (g $100mL^{-1}$ blood)	15.1	9.4	13.8 - 17.2 *(Adult Males)* / 12.1 - 15.1 *(Adult females)*

Following the investigations, the medical team proposed the following interventions for improving the health of the residents:

(i) Clean and dress wounds immediately after injury.

(ii) Encouraged regular intake of iron-rich foods.
(iii) Encouraged people working in hot conditions, to regularly drink sugary juices.

The residents need to know when to implement the proposed interventions.
Task:

(a) Analyse the relationship between the health challenges experienced by the residents and the medical findings.
(b) Develop a justified implementation plan showing when each proposed intervention should be applied.

Item 4

An outbreak of influenza occurred at a secondary school during a cold season. James was among the students who contracted the disease and recovered after receiving supportive treatment at the school sick bay. Six months later, another influenza outbreak occurred in the same school. Although James had close contact with infected classmates, he did not develop any symptoms whereas several students who had never suffered from influenza became ill.

During the outbreak, Joan who experienced a moderate fever recovered quickly than Brian who experienced a high fever. The school invited a doctor to investigate the differences in recovery among students. Blood samples were collected from James, Peter, Joan and Brian for laboratory analysis. Also, body temperature, blood flow to the skin and metabolic rate were determined.

Table 6: Shows results for the laboratory analysis of blood samples and other physiological factors.

Table 6

Physiological factor	**James**	**Joan**	**Brian**
Contraction of infection first time	Yes	NO	NO
Memory lymphocyte cells/mm³	4,800	400	410
Metabolic rate (KJ/day)	7,100	7,200	9,500
Blood flow to skin (ML/min)	250	3,000	6,000
Body temperature	37.0	38.2	39.8

The Doctor explained that during an immune response, the energy demand of cells raises to increase production of lymphocytes and also activate other immune cells.
Basing on the effects of influenza virus on body functioning, the doctor recommended

the school administration to use interventions that would increase number of memory lymphocyte, curb the spread of infection and improve energy production.

Task:

(a) Analyse the differences in the response of the students body to influenza virus and also explain how the infection affected the physiology state of their bodies.

(b) Evaluate interventions that the school administration can use to address the doctor's recommendations.

SECTION B:

Part II

Respond to **one** *item from this Part*

Item 5

A wetland ecosystem was restored after five years of degradation by introducing native aquatic plants and fish. During the first four years, populations of most organisms recovered, energy flow was efficient, species diversity was high and the ecosystem remained stable.

Later, farmers repeatedly sprayed the same pesticide to control insect pests on a crop farm that was close to the wetland. During the rainy season, some of the pesticide entered the wetland. Several dead fish were observed shortly after spraying. After several years, the pesticide became less effective against the insect pests. Ten years later, prolonged drought reduced the water levels in wetland into isolated pools and some aquatic organisms were no longer found. A survey conducted by the environmental conservationists revealed changes in species composition and abundance within the wetland.

Table 7 shows a record of changes in population size of selected wetland organisms.

Organism	Population after restoration	Population after 10 years
Aquatic Plants	8,600	12,900
Frogs	780	620
Fish	1,450	260
Fish-eating birds	210	48

Table 8 shows a record of the concentration of pesticide in selected components of the wetland.

Component	Pesticide concentration (ppm)
Water	0.002
Aquatic Plants	0.04
Frogs	0.28
Fish	1.60
Fish-eating birds	7.80

Following the observations made, environmental conservationists recommended reducing dependence on pesticides, adopting Integrated Pest Management (IPM) and gazetting the wetlands. However, the farmers wanted to evaluate the proposed measures before deciding which ones to adopt.

Task:

(a) Analyze the changes in species composition, abundance of organisms in the wetland and also reduction in effectiveness of the pesticide used on the crop farm.

(b) Evaluate the effectiveness of the proposed measures and recommend to farmers the most appropriate measure(s) for promoting ecosystem stability.

Item 6

A traditional herbal medicine company introduced an exotic shrub on land adjacent to a game reserve as a source of raw materials for making herbal medicines. Several years later, the shrub spread rapidly into the game reserve where it formed dense thickets that replaced native grasses and shrubs. To determine the impact of the exotic shrub on plant and animal populations in the game reserve, the wild life authority conducted a study. Table 7 shows a record of results from the study.

Table 7.

Parameter	Before Invasion	After Invasion
Area covered by native grasses (ha)	640	210
Area covered by exotic shrubs (ha)	15	390
Estimated carrying capacity for Antelopes	1500	650
Antelope Population	1,180	540
Lion Population	96	42

During the same period, the authority established breeding enclosures to increase the declining antelope population. Only males with large body size and long horns were allowed to mate with the females because these characteristics were considered desirable for future breeding programmes. Eight years later, the environmentalists determined the allele frequency of allele **A** and **a** in the antelope population. Table 8 shows the results.

Table 8

Period	**Frequency of Allele A (p)**	**Frequency of allele a (q)**
Before selective breeding	0.62	0.38
Eight years after selective breeding	0.79	0.21

Allele A codes for production of a skin coat colour that provides camouflage against predators.

Following the study, the wildlife authority consulted an environmental expert on appropriate management strategies. The expert recommended allowing random mating during captive breeding of antelopes, using appropriate herbicides to control the exotic shrub, and close monitoring of antelope population. However, the authority wanted to choose one recommendation that would solve their challenges.

Task:

(a) Analyze the causes of changes in populations of plant and animals in the game reserve and also changes in allele frequency.

(b) Defend the environmental expert's recommendation and advise the wildlife authority accordingly.

END